If You Take The Road Less Travelled by, Does it Make a Difference?

Pittsburgh Penguins MSP Consulting Project

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Agenda

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Introduction: Overview

- There are multiple traditional paths hockey prospects can take to get to the NHL:
  - USHL → NCAA → NHL
  - USHL (→ NCAA) → AHL → NHL
  - International → KHL → NHL
  - Other defined paths
- Most players do not immediately go to the NHL when they are eligible (drafted or not). They stay in or move to some “development leagues” before entering the NHL.
  - Draft eligibility (North America): Players must be 18 years old by 15 September or under 20 years old by 31 December in the year of the draft.
  - Development leagues: USHL, NCAA, etc.

Introduction: Research Question

- Questions: Does taking different development paths matter? How do players' development paths impact their performance and success in the NHL?
- The understanding in the scouting community is that development path does matter.
  - Only anecdotal.
  - We intend to establish grounding on this thought.

Introduction: Overview

- People have very strong opinions about how players' development paths impact their future in the NHL.
  - Typically, American players who take the NCAA path have higher success rates (e.g. 20% make the NHL, compared to 5% from the USHL path)
  - However, the NCAA player pool are already better in terms of quality. Better players are getting their opportunities in the NCAA.
  - Is there causal impact of taking the NCAA path?

Data

- Two datasets:
  - Leagues: NHL, NCAA, USHL and AHL
  - Time period: 2001 - 2020
    - *contains some data earlier than 2001*
  - Players' biographical information
  - Players' performance data each season
    - *box score statistics*

Data Description

- Biographical information:
  - 15786 players

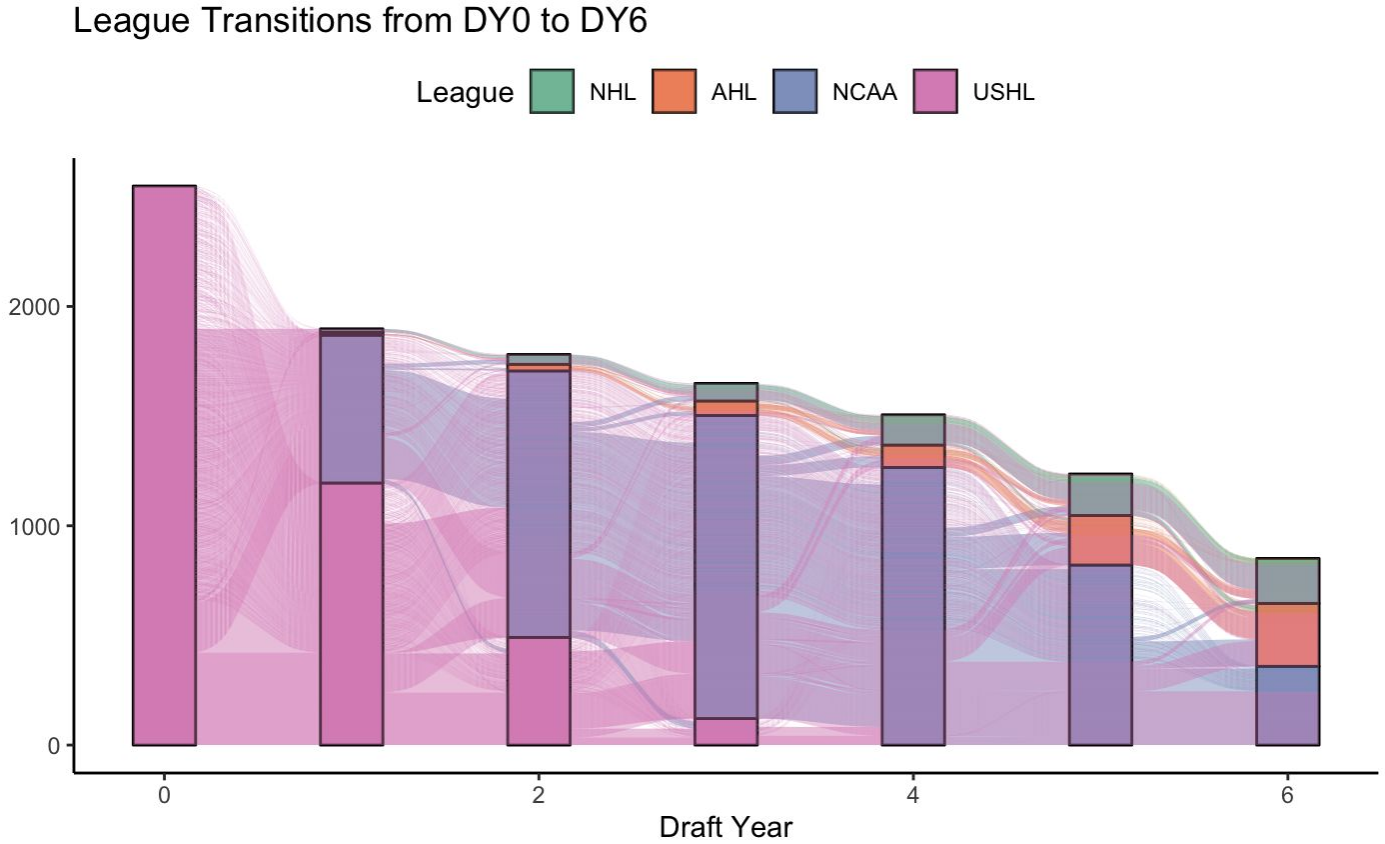
	Player	Position	DateofBirth	Height	Weight	Nation	Shoots
1	Scott May	C	Jan 08, 1982	5'10" / 178 cm	187 lbs / 85 kg	Canada	R
2	Kent Gillings	F	Jun 14, 1979	5'10" / 177 cm	194 lbs / 88 kg	Canada / Ireland	R
3	Tyler Kindle	D	Feb 20, 1978	5'8" / 173 cm	165 lbs / 75 kg	USA	L
4	D'Arcy McConvey	C	Oct 23, 1981	5'10" / 177 cm	185 lbs / 84 kg	Canada	L
5	Lloyd Marks	C	Oct 21, 1977	5'8" / 173 cm	174 lbs / 79 kg	Canada	L
6	Jason Deskins	C	May 06, 1979	5'10" / 178 cm	185 lbs / 84 kg	USA	-
7	Jim Abbott	LW/C	May 03, 1980	6'1" / 186 cm	185 lbs / 84 kg	USA	L

Data Description

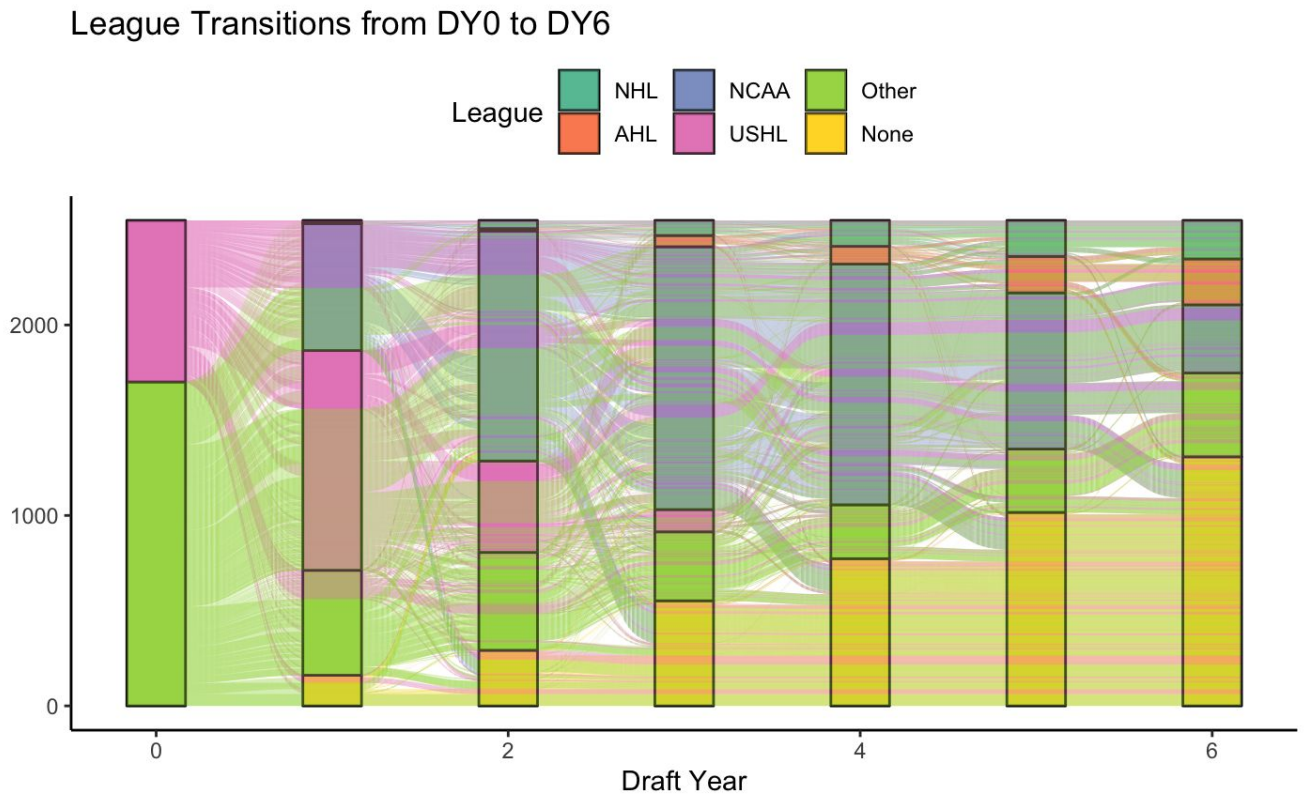
- Player performance:
  - 266326 rows (15220 of players * # of seasons they played)

	Player	Season	Team	League	Games	Goals	Assists	TotalPoints	PenaltyMinutes	PlusMinus
1	Scott May	1998-99	South Surrey Eagles	BCHL	45	10	28	38	23	
2	Scott May	1999-00	South Surrey Eagles	BCHL	54	42	42	84	86	
3	Scott May	2000-01	Ohio State Univ.	NCAA	37	9	9	18	26	-3
4	Scott May	2001-02	Ohio State Univ.	NCAA	40	12	17	29	42	4
5	Scott May	2002-03	Ohio State Univ.	NCAA	43	10	25	35	56	5
6	Scott May	2003-04	Ohio State Univ.	NCAA	41	15	19	34	42	4
7	Scott May		St. John's Maple Leafs	AHL	5	1	1	2	2	3
8	Scott May	2004-05	St. John's Maple Leafs	AHL	16	0	1	1	21	-3

EDA


League Transitions from DY0 to DY6
League
NHL
AHL
NCAA
USHL
2000
1000
0
0
2
4
6
Draft Year

EDA



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Method: Causal Inference

- Goal: determine the causal effect of development paths (Treatment **Z**) on players' future in the NHL (Response **Y**), controlling for player quality etc. (Confounders **X**)
  - Conditional Average Treatment Effect (CATE):

$$E[Y \mid Z = z1, X] - E[Y \mid Z = z0, X]$$

  - The fundamental problem: our samples are biased (e.g. better prospects are more likely to enter NCAA than USHL)

Method: Two solutions

- Solution 1: Control the treatment assignment mechanism; then estimate causal effect just as in a randomized experiment
  - Method: Propensity Score Weighting
- Solution 2: If we can precisely estimate a model for the outcome $Y = f(z, x) + \varepsilon$, then we can calculate CATE
  - Method: Bayesian Additive Regression Trees (BART)

Method

Propensity Score Weighting

- Used to reweight the data so that we don't have any selection effect or bias in our treatment.
- We use logistic regression to predict the treatment T as well as possible from all of the predictors.
  - P(T = NCAA | X) is our **propensity score.** $\hat{e}(x) = P(T = NCAA|X)$
  - Reweight the original data based on the propensity score. We explored two weighting methods:

Method 1

For treatment group

$$\omega = \frac{1}{1 - \hat{e}(x)}$$

For control group

$$\omega = \frac{1}{\hat{e}(x)}$$

Method 2

$$\omega = \hat{e}(x)(1 - \hat{e}(x))$$

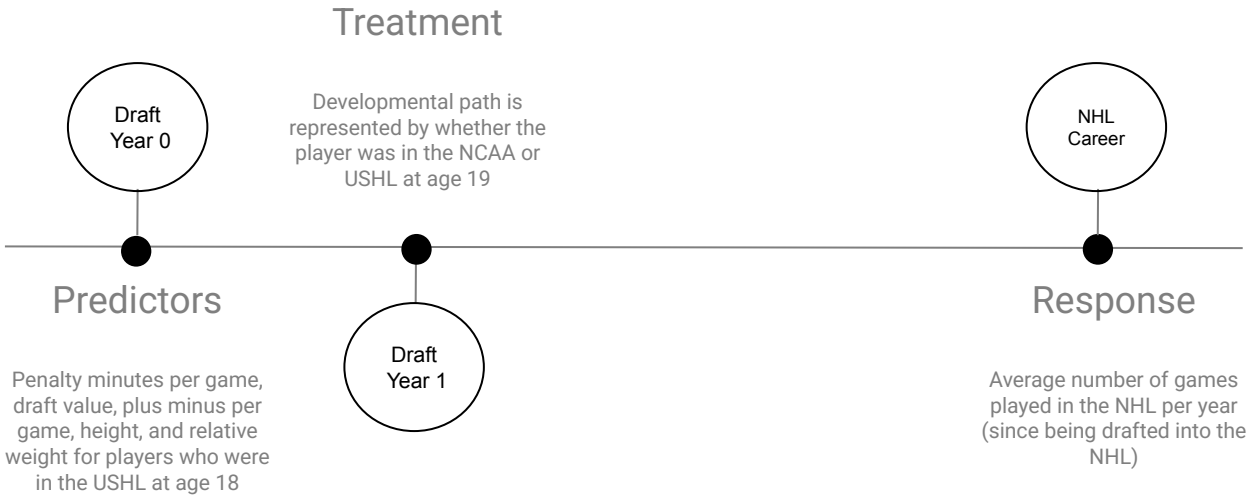
- A potential drawback of propensity scores when used for matching is that a very large number of subjects may be needed
- Propensity scores can also be used as weights in a linear model such as regression or ANOVA, so all the subjects in the control and treatment group can be used for this application

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Method

Propensity Score Weighting

What is the causal effect of developmental path on player success in the NHL?



Forward Positioned Players

1. Obtain the propensity score by modeling P(T = NCAA) based on predictors in draft year 0
2. Predict success in NHL with propensity score weights on the data
3. Interpret coefficient estimate for the treatment variable

Defensemen

1. Obtain the propensity score by modeling P(T = NCAA) based on predictors in draft year 0
2. Predict success in NHL with propensity score weights on the data
3. Interpret coefficient estimate for the treatment variable

Method

BART

- Estimate Y = f (z, x) + ε, where ε ~ N(0, σ^2) using a **sum-of-trees** model
- The idea is to fit a bunch of **weak-learning** (small) trees, each fitting to the residuals of the previous trees. Then, **additively** combine these trees to reduce bias, similar to boosting.
- Introduce a **regularization prior** to avoid overfitting. The prior controls the size of the trees (T), the magnitude of the outputs of the trees (M), and the value of σ.
- Compute the **posterior** using **Markov Chain Monte Carlo** (MCMC)
- At each iteration of MCMC, (T, M) and σ are redrawn to seek a good f.
- After estimating Y using BART, we can calculate CATE by

$$\frac{1}{n}\sum_{i=1}^{n} f(1, x_i) - f(0, x_i)$$

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Method

Bayesian Additive Regression Trees (BART)

- Selected 1830 Forward and 996 Backward players who played in some developmental league in DY

- Predictors:
  - Games, Goals, Assists, PenaltyMinutes, Position, Height_cm, Weight_kg

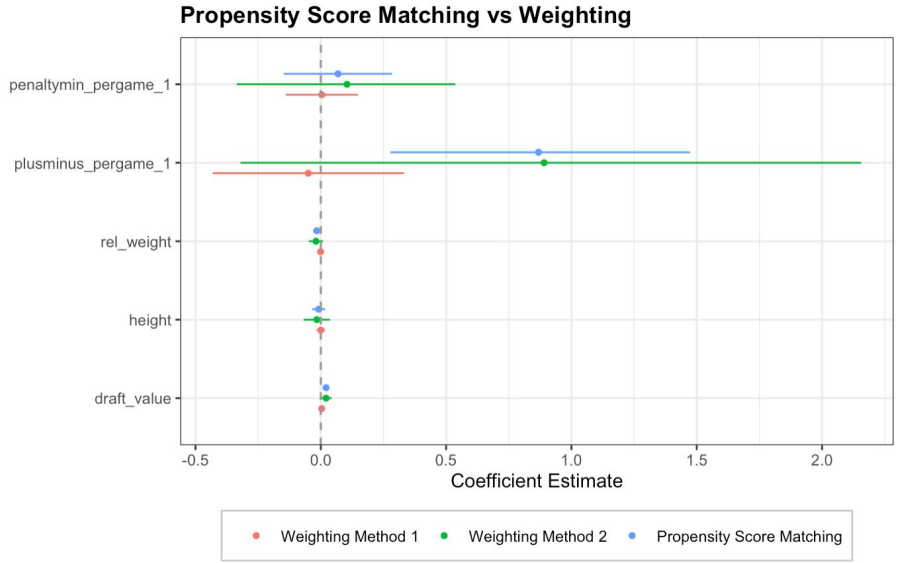
- Response variable:
  - The average number of games played in the NHL per season

- After obtaining the model: predict performance with {USHL, NCAA} x {Defense, Forward} on whole dataset

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Results

Propensity Score Weighting vs Matching : Forward Players



glm(NCAA ~ penaltymin_pergame_1 + plusminus_pergame_1 +rel_weight + height + draft_value , family = 'binomial'))

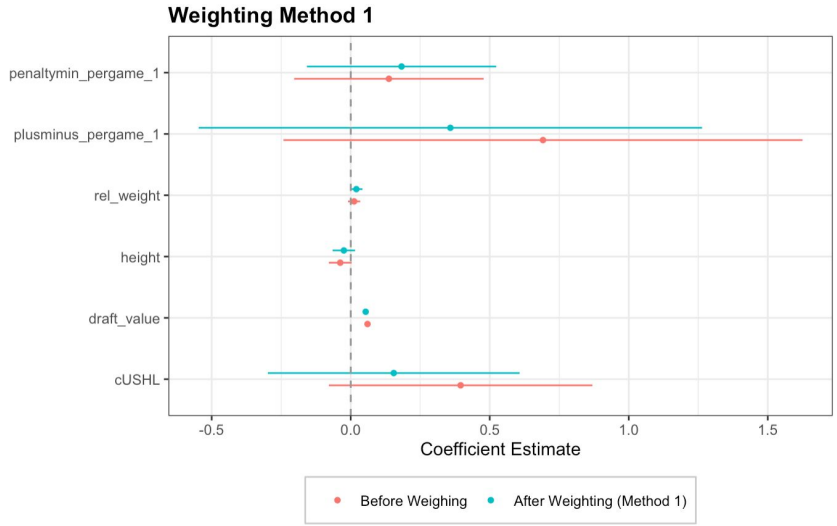
- We have **removed selection bias if the coefficient estimates are zero** when predicting treatment effect
  - This indicates that the confounders no longer have an effect on the treatment assignments
- Weighting Method 1 has 95% CIs that contain zero in all coefficient estimates
- Removed goals and assists per game due to correlation with draft value
- Rel_weight is player weight after adjusting for its relationship with height

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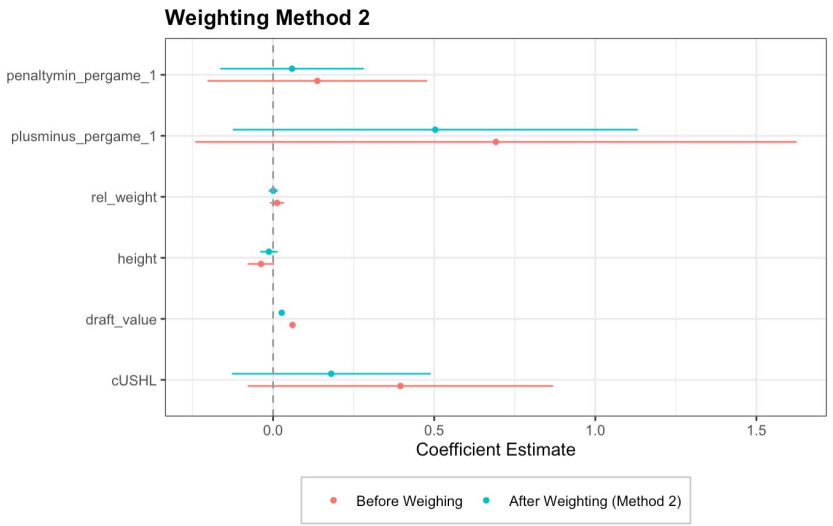
Results

Propensity Score Weighting: Forward Positioned Players

Predicting Success in the NHL: Weighting Method 1



Predicting Success in NHL: Weighting Method 2

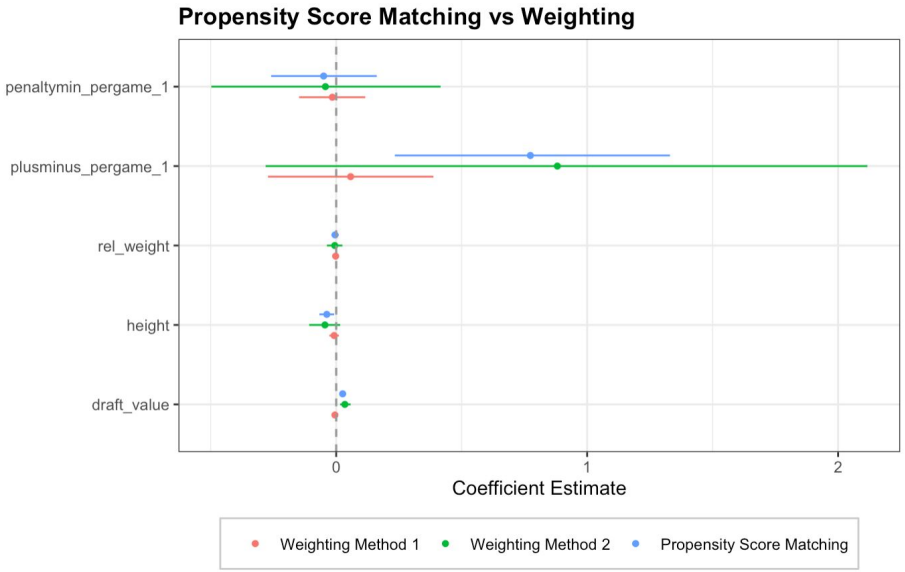


- After addressing for selection bias using both methods, we observe that developmental league is not a significant predictor for success in the NHL
- Draft value appears to be a significant predictor after propensity score weighting

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Results

Propensity Score Weighting vs Matching : Defensemen



glm(NCAA ~ penaltymin_pergame_1 + plusminus_pergame_1 +rel_weight + height + draft_value , family = 'binomial')

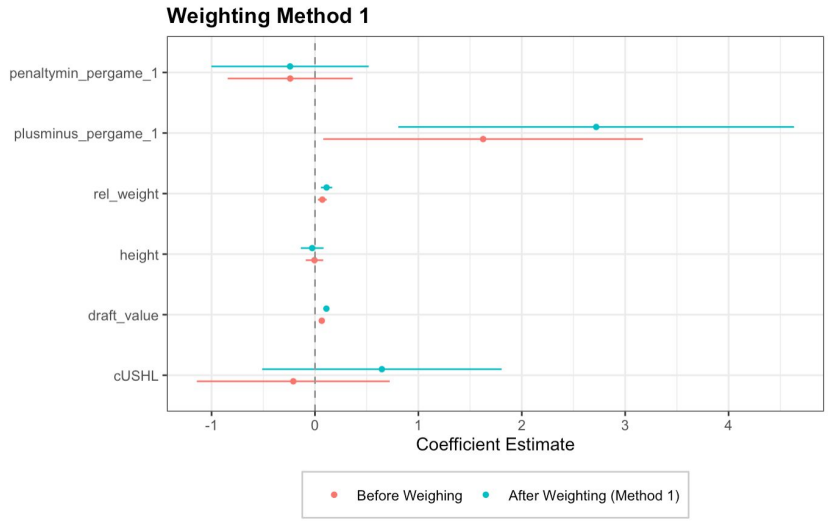
- We have **removed selection bias if the coefficient estimates are zero** when predicting treatment effect
  - This indicates that the confounders no longer have an effect on the treatment assignments
- Weighting Method 1 has 95% CIs that contain zero in all coefficient estimates
- Removed goals and assists per game due to correlation with draft value
- Rel_weight is player weight after adjusting for its relationship with height

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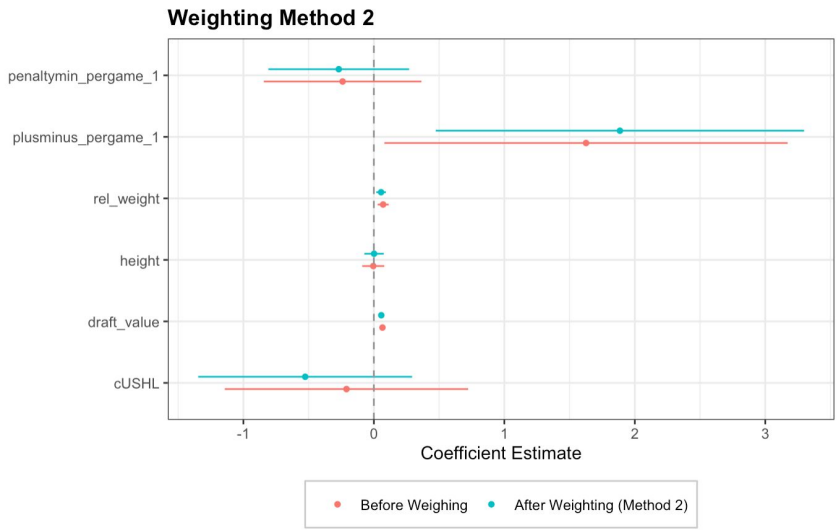
Results

Propensity Score Matching: Defensemen

Weighting Method 1



Weighting Method 2



- After addressing for selection bias using both methods, we observe that developmental league is not a significant predictor for success in the NHL
- Draft value, relative weight, and plus minus appears to be significant predictors after propensity score weighting

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Result -- Treatment Effects (NCAA vs. USHL)

BART

Y = The average number of games played in the NHL per season

	Observed Average Difference	CATE
Forward	13.36513	4.64768
Defense*	14.65782	4.63431

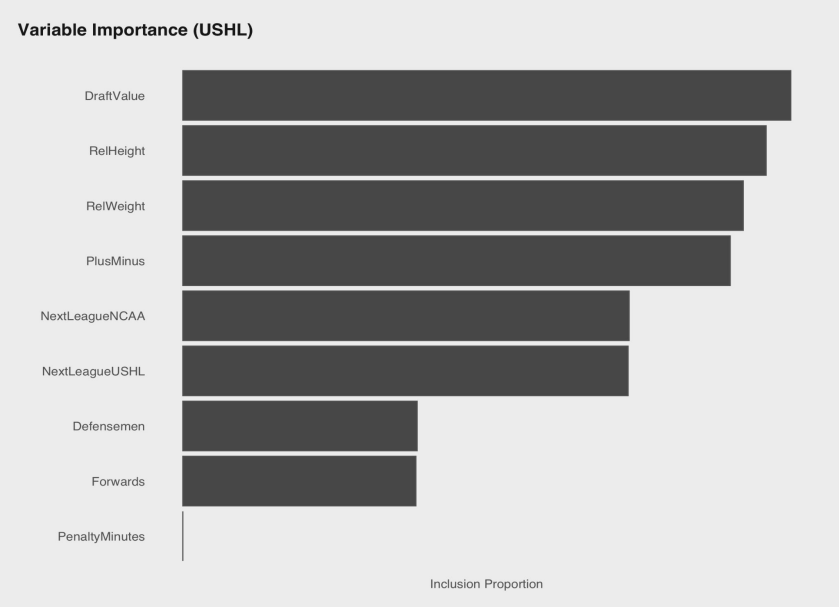
*Defense: "D/F", "D/LW", "D"

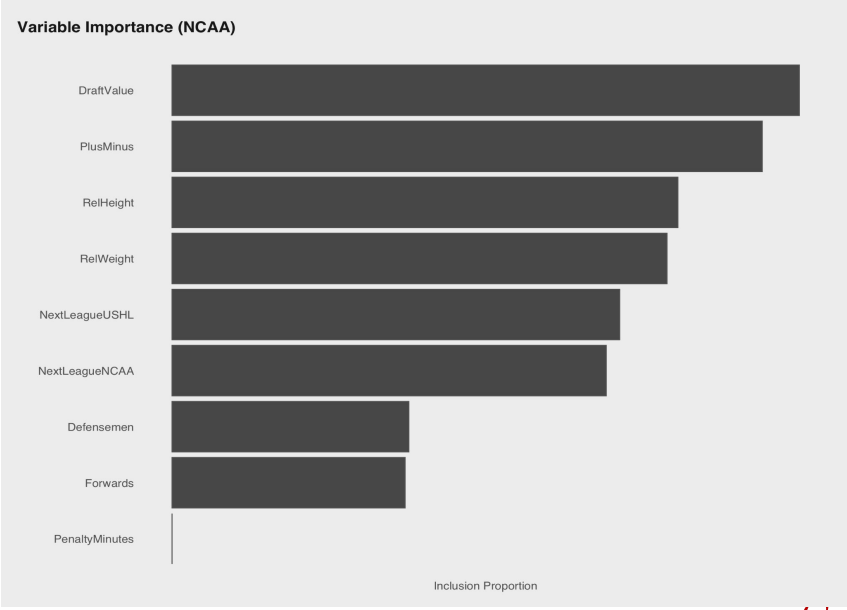
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Result

BART

Variable Importance, for players in USHL vs. NCAA in their first draft year





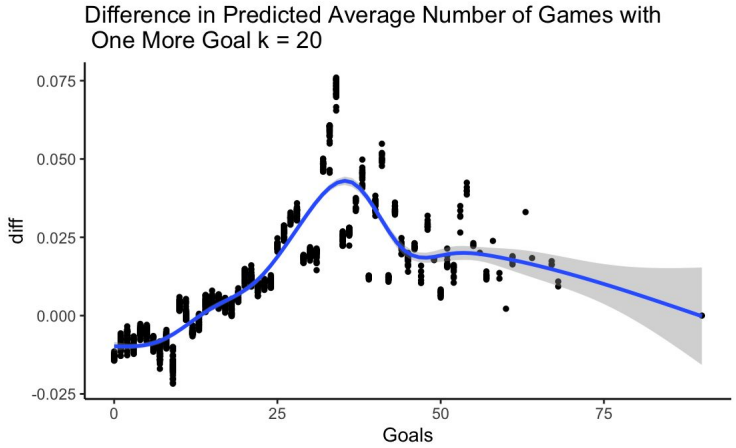
What if ...

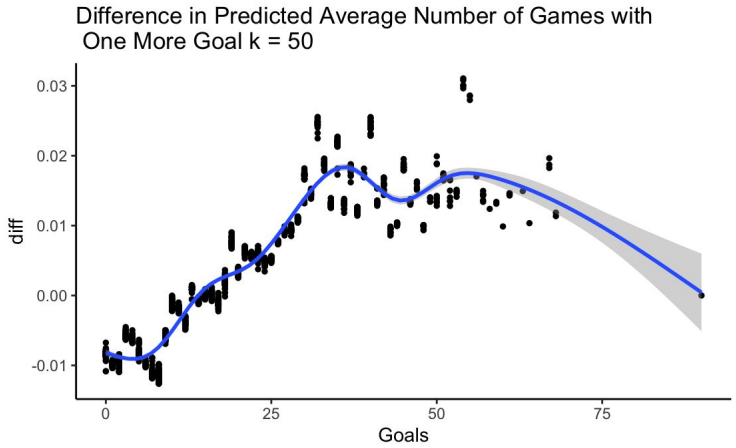
I have one more goal?

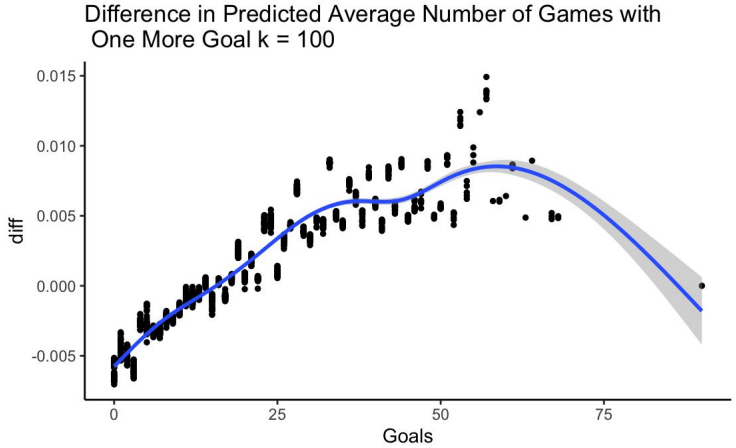
I have one more assist?

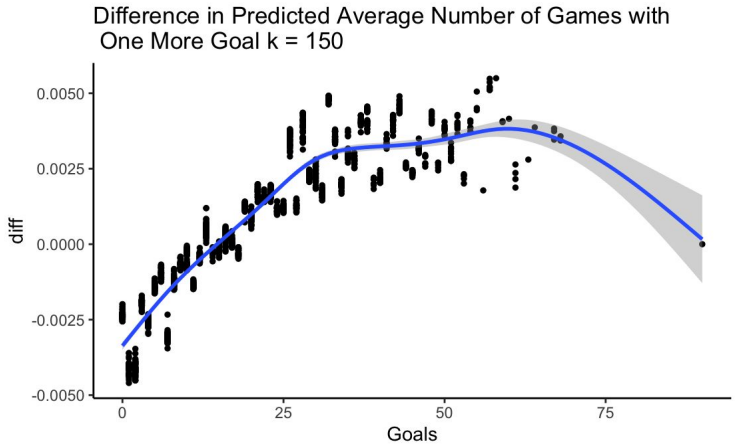
I am 1cm/0.9inch taller?

I am 1kg/2.2pounds heavier?







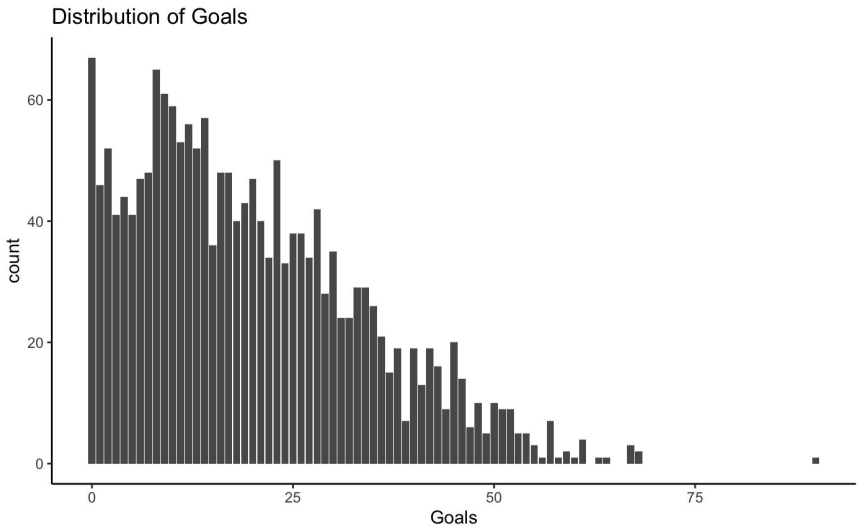


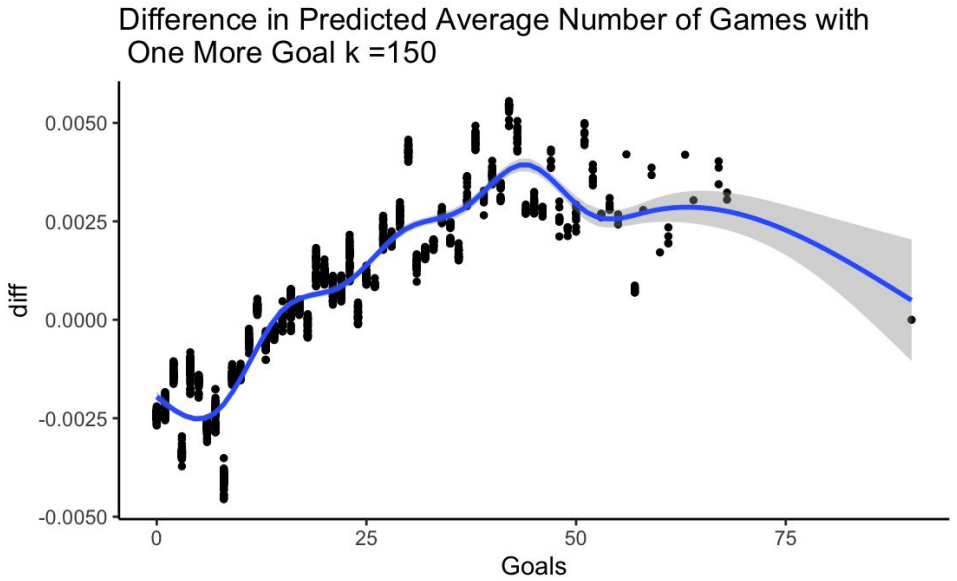
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Result

BART

- What if one more goal for forward players?



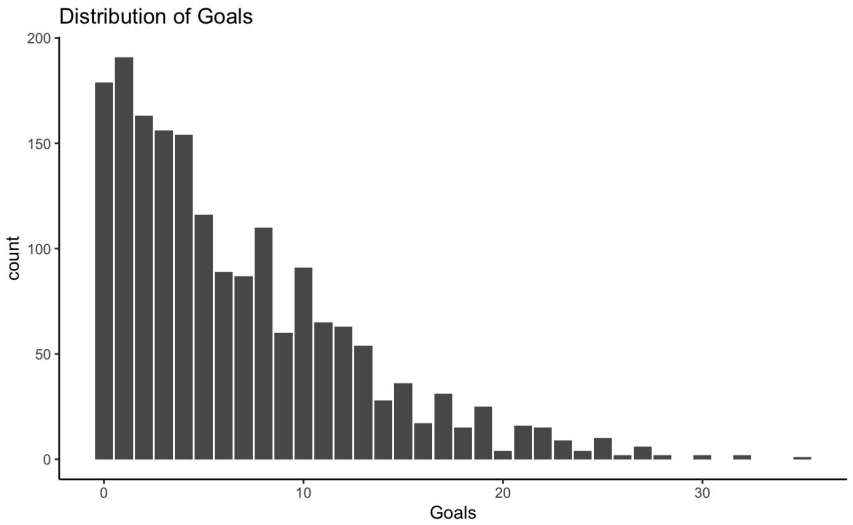


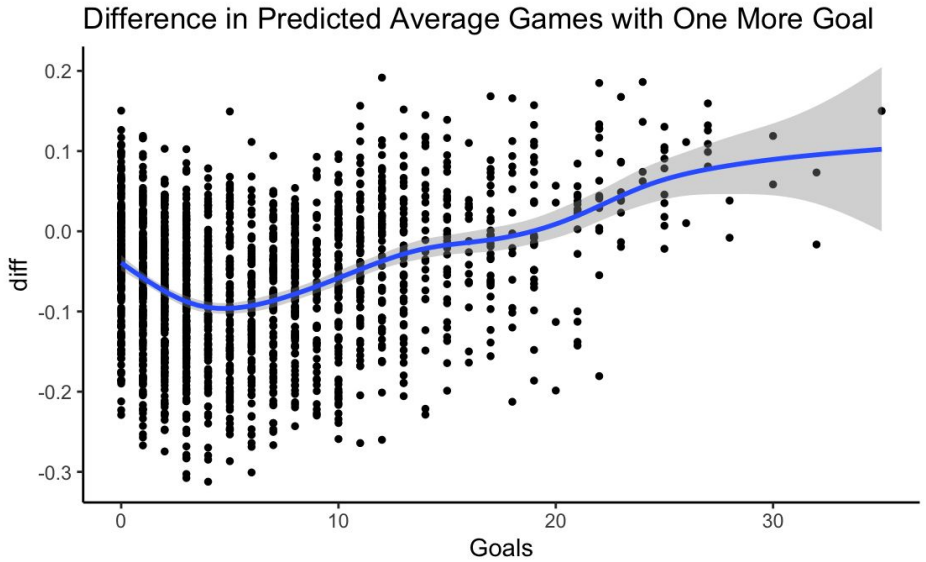
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Result

BART

- What if one more goal for defense players?



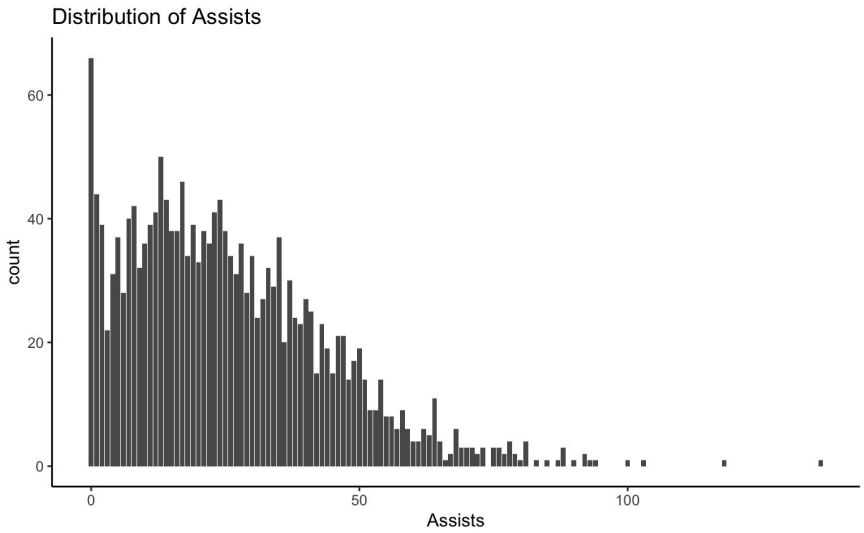


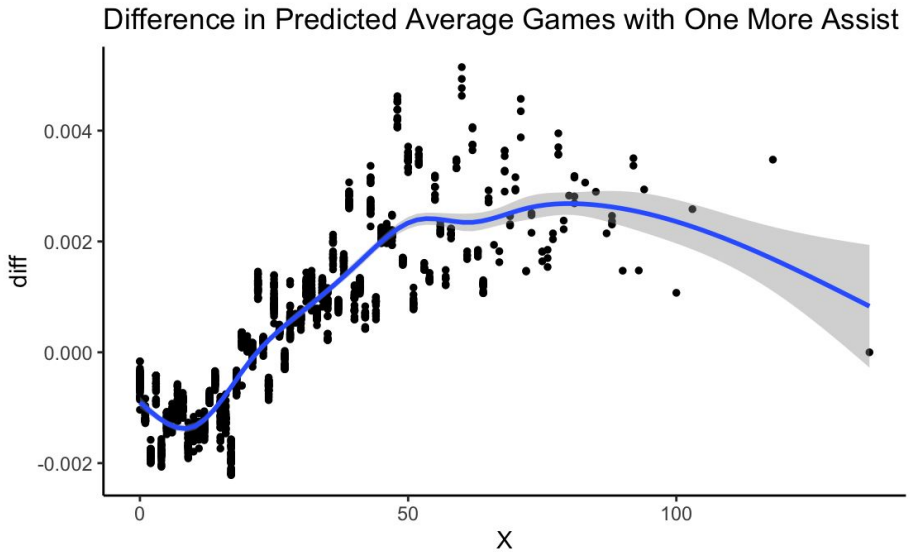
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Result

BART

- What if one more assist for forward players?



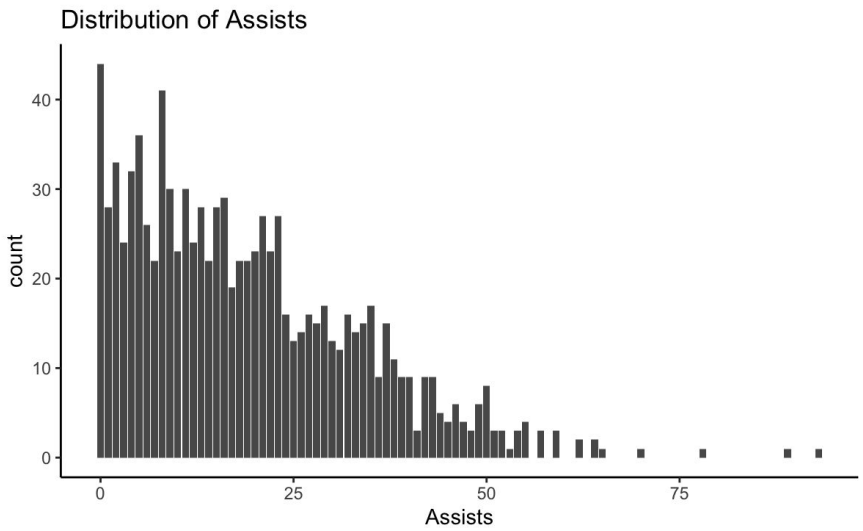


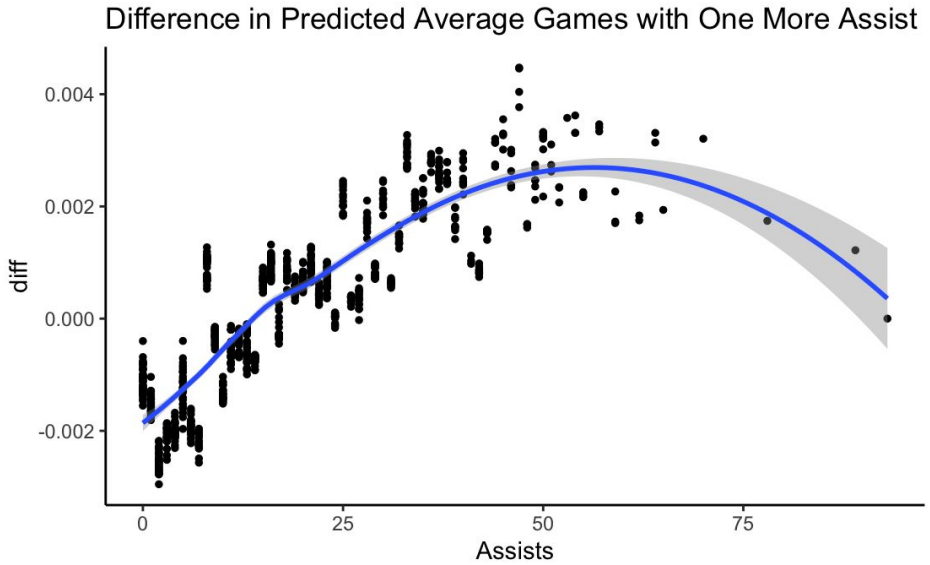
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Result

BART

- What if one more assist for defense players?



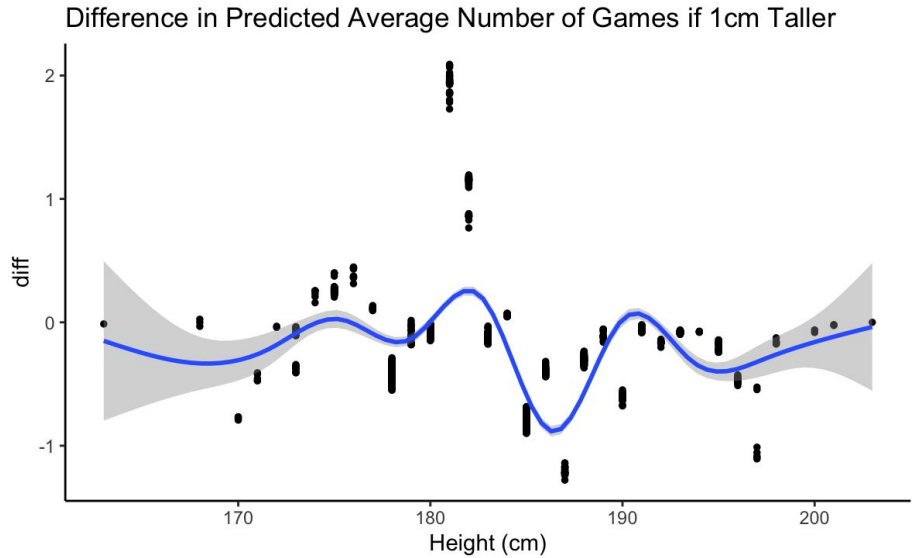


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Result

BART

- What if one centimeter/(0.39 inches) taller for forward players?



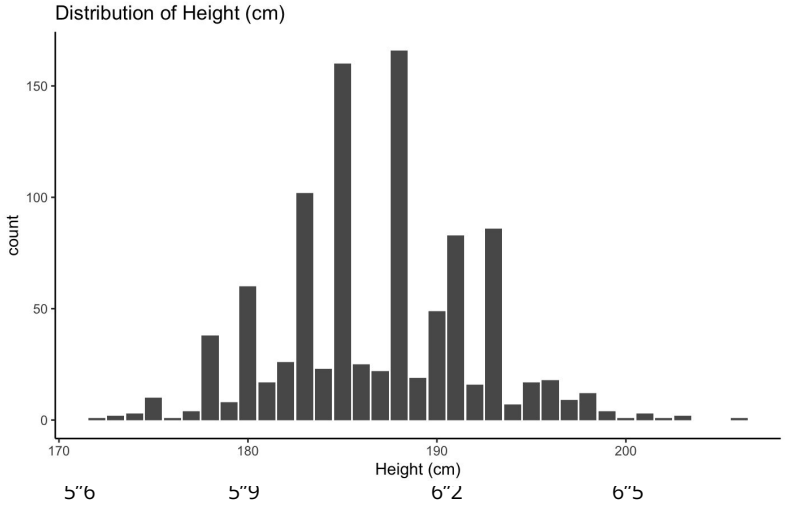
5″6 5″9 6″2 6″5

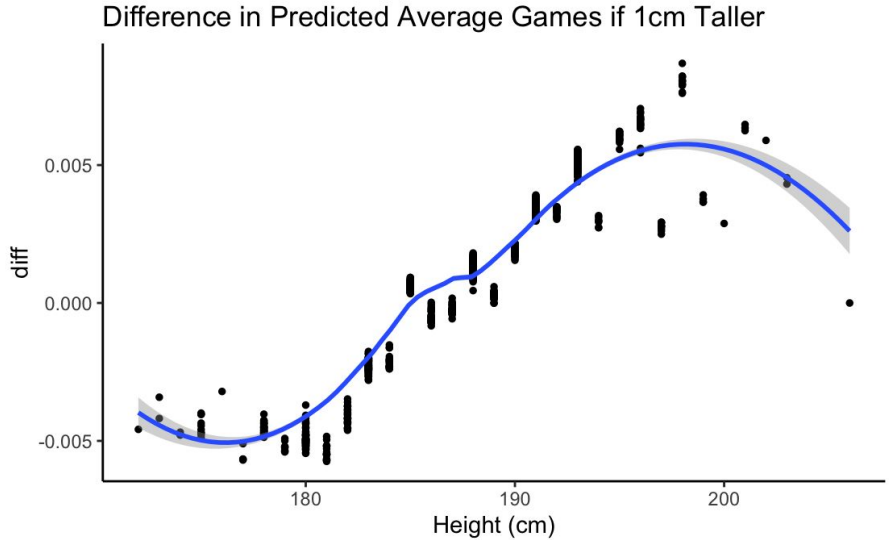
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Result

BART

- What if one centimeter/(0.39 inches) taller for defense players?



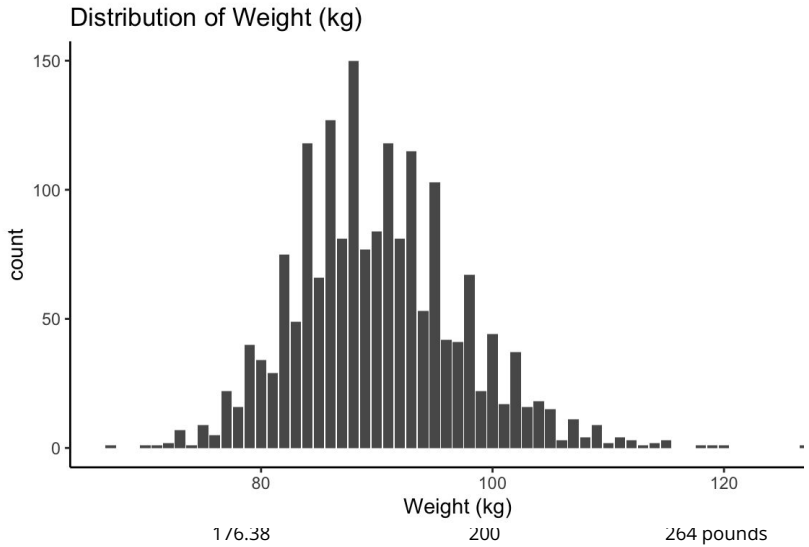


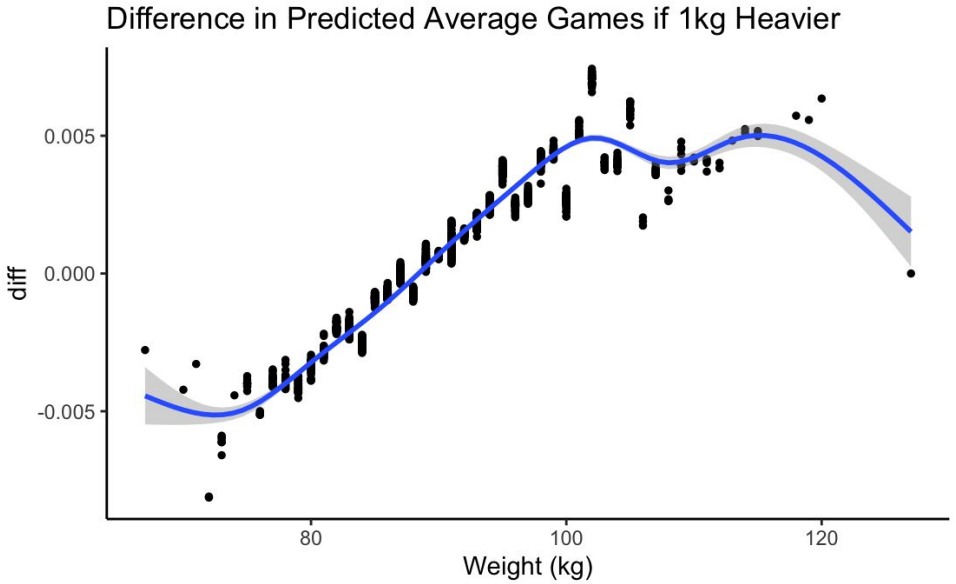
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Result

BART

- What if 1kg/2.2 pounds heavier for forward players?



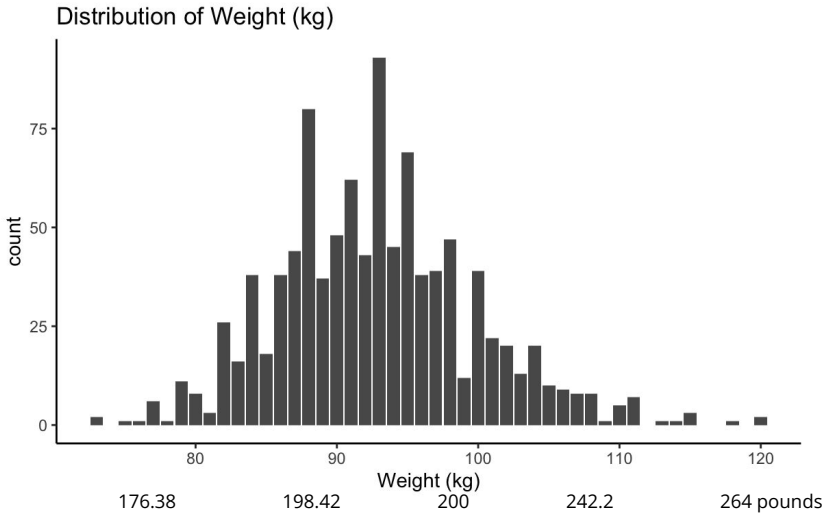


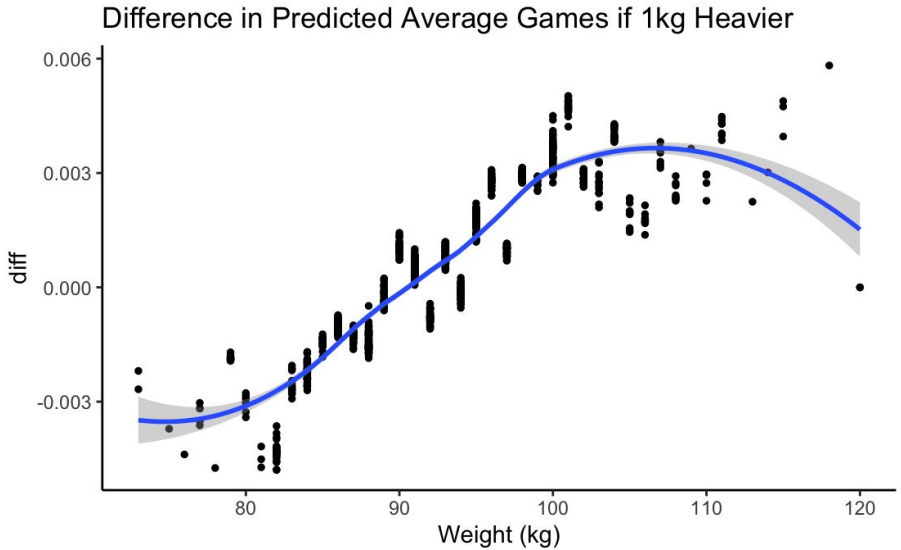
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Result

BART

- What if one 1kg/2.2 pounds for defense players?





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Q&A

Appendix

Thank You!